Light

GCSE Computer Science — CPU Architecture (Set 1)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset

← Back to Cpu Architecture

Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is the main purpose of the Central Processing Unit (CPU)?

- ○ **A.** To fetch and execute instructions stored in memory
- ○ **B.** To store files permanently when the computer is switched off
- ○ **C.** To connect the computer to a network
- ○ **D.** To display images on the monitor

Q2 (1 mark)

Which component of the CPU performs arithmetic and logical operations?

- ○ **A.** Control Unit

- ○ **B.** Arithmetic Logic Unit
- ○ **C.** Program Counter
- ○ **D.** Memory Data Register

Q3 (1 mark)

What is the role of the Control Unit (CU) in the CPU?

- ○ **A.** To perform mathematical calculations
- ○ **B.** To store the results of calculations
- ○ **C.** To coordinate and manage the execution of instructions
- ○ **D.** To connect the CPU to main memory

Q4 (1 mark)

What type of memory is cache?

- ○ **A.** Very fast memory built into or close to the CPU
- ○ **B.** Secondary storage that holds files permanently
- ○ **C.** Memory used only for storing the operating system
- ○ **D.** External memory connected via USB

Q5 (1 mark)

What are registers in a CPU?

- ○ **A.** Large storage areas for saving documents
- ○ **B.** Devices that connect the CPU to peripherals
- ○ **C.** Small, extremely fast storage locations within the CPU
- ○ **D.** Software programs that manage the CPU

Q6 (1 mark)

In Von Neumann architecture, where are both programs and data stored?

- ○ **A.** In separate memory units
- ○ **B.** In the same memory unit
- ○ **C.** Programs in cache, data in RAM
- ○ **D.** Programs on the hard drive only

Q7 (1 mark)

What does the Memory Address Register (MAR) hold?

- ○ **A.** The address of a memory location to be accessed
- ○ **B.** The data being read from or written to memory
- ○ **C.** The next instruction to be executed
- ○ **D.** The result of the most recent calculation

Q8 (1 mark)

What is the function of the Memory Data Register (MDR)?

- ○ **A.** To hold the address of the next instruction
- ○ **B.** To count how many instructions have been executed
- ○ **C.** To hold data that is being transferred to or from memory
- ○ **D.** To decode instructions into control signals

Q9 (1 mark)

What does the Program Counter (PC) store?

- ○ **A.** The total number of programs installed
- ○ **B.** The address of the next instruction to be fetched
- ○ **C.** The result of the current calculation
- ○ **D.** A copy of all data in RAM

Q10 (1 mark)

What is the purpose of the Accumulator register?

- ○ **A.** To hold intermediate results of calculations performed by the ALU
- ○ **B.** To store the address of the current instruction
- ○ **C.** To manage connections to external devices

- ○ **D.** To store the operating system

Q11 (1 mark)

What does clock speed measure in a CPU?

- ○ **A.** The number of programs that can run simultaneously
- ○ **B.** The amount of data that can be stored
- ○ **C.** The number of cycles the CPU can perform per second
- ○ **D.** The physical size of the processor

Q12 (1 mark)

What is a CPU core?

- ○ **A.** The cooling fan attached to the processor
- ○ **B.** An independent processing unit within the CPU
- ○ **C.** The socket where the CPU connects to the motherboard
- ○ **D.** The brand name of the processor

Q13 (2 marks)

A computer has a CPU with 8MB of cache memory.
Explain why having more cache memory can improve the performance of a CPU.

- ○ **A.** More cache means more permanent storage for files
- ○ **B.** More cache increases the clock speed of the processor
- ○ **C.** More cache allows more programs to be installed on the computer
- ○ **D.** More cache allows more frequently used data and instructions to be stored close to the CPU, reducing the time spent accessing slower main memory

Q14 (2 marks)

A user upgrades from a dual-core processor to a quad-core processor. Which statement best explains how this affects performance?

- ○ **A.** The CPU can execute more tasks simultaneously, improving performance for multi-threaded applications
- ○ **B.** The computer can now store four times as much data
- ○ **C.** Each individual program will run exactly twice as fast
- ○ **D.** The clock speed of the processor will automatically double

Q15 (2 marks)

During a memory read operation, what is the relationship between the MAR and MDR?

- ○ **A.** The MAR and MDR both store the same data simultaneously
- ○ **B.** The MDR holds the address, and the MAR receives the data
- ○ **C.** The MAR holds the address to access, and the MDR receives the data from that address

- ○ **D.** The MAR and MDR are both used to store calculation results

Q16 (2 marks)

The Von Neumann architecture has a limitation known as the 'Von Neumann bottleneck'.
What causes this bottleneck?

- ○ **A.** The CPU cannot perform calculations quickly enough
- ○ **B.** Programs and data share the same memory and bus, limiting how fast data can be transferred
- ○ **C.** Modern software requires too much storage space
- ○ **D.** The Control Unit cannot decode instructions fast enough

Q17 (2 marks)

A CPU has a clock speed of 3.5 GHz.
What does this specification indicate about the processor?

- ○ **A.** The CPU has 3.5 billion transistors
- ○ **B.** The CPU has 3.5 gigabytes of cache
- ○ **C.** The CPU can run 3.5 billion programs simultaneously
- ○ **D.** The CPU can perform 3.5 billion cycles per second

Q18 (2 marks)

A programmer writes code that compares two values to check if they are equal.
Which CPU component performs this comparison?

- ○ **A.** The Arithmetic Logic Unit, which performs logical comparisons
- ○ **B.** The Control Unit, which manages all operations
- ○ **C.** The Program Counter, which tracks instruction addresses
- ○ **D.** The Memory Data Register, which holds data from memory

Q19 (2 marks)

Modern CPUs typically have multiple levels of cache (L1, L2, L3).
Which statement correctly describes these cache levels?

- ○ **A.** L3 cache is the fastest because it is the largest
- ○ **B.** All cache levels have the same speed but different capacities
- ○ **C.** L1 cache is the smallest and fastest, located closest to the CPU cores
- ○ **D.** L1 cache is the largest and holds the most data

Q20 (2 marks)

During the execution of an addition instruction, data flows through several CPU components.
Which sequence correctly shows the flow of data for this operation?

- ○ **A.** ALU → Accumulator → Control Unit → RAM
- ○ **B.** Accumulator → Control Unit → ALU → MDR
- ○ **C.** RAM → Control Unit → Accumulator → ALU
- ○ **D.** Control Unit → RAM → MDR → ALU → Accumulator

Q21 (2 marks)

A program contains an IF statement that causes the code to jump to a different location.
How is the Program Counter (PC) affected when a branch instruction is executed?

- ○ **A.** The PC is reset to zero
- ○ **B.** The PC continues to increment normally
- ○ **C.** The PC is cleared and the program ends
- ○ **D.** The PC is loaded with the address of the branch target

Q22 (2 marks)

Two CPUs have the following specifications:
CPU A: 4 cores, 3.0 GHz, 8MB cache
CPU B: 2 cores, 4.0 GHz, 4MB cache
Which statement best compares these processors?

- ○ **A.** CPU A will always be faster because it has more cores
- ○ **B.** CPU A may perform better for multi-threaded tasks, while CPU B may perform better for single-threaded tasks
- ○ **C.** CPU B will always be faster because it has higher clock speed
- ○ **D.** Both CPUs will perform identically because their specifications balance out

Q23 (2 marks)

When the CPU requests data from memory, it first checks the cache.
What happens during a 'cache miss'?

- ○ **A.** The requested data is found in the cache and returned immediately
- ○ **B.** The cache is cleared and reset
- ○ **C.** The program crashes due to missing data
- ○ **D.** The CPU must wait while the data is fetched from slower main memory

Q24 (3 marks)

A computer is executing a program that calculates the average of a list of numbers.
Explain how the Control Unit, ALU, and Accumulator work together to perform this calculation.

- ○ **A.** The Control Unit fetches numbers from memory, the ALU adds them and divides by the count, and the Accumulator holds intermediate sums and the final average
- ○ **B.** The Control Unit calculates the sum, the ALU stores the result, and the Accumulator manages memory
- ○ **C.** The Accumulator fetches instructions, the Control Unit performs addition, and the ALU stores all data
- ○ **D.** The ALU fetches numbers, the Accumulator performs division, and the Control Unit stores the result

Q25 (3 marks)

During the fetch-decode-execute cycle, the CPU uses several registers. Describe the role of the Program Counter (PC), Memory Address Register (MAR), and Memory Data Register (MDR) during the fetch stage.

- ○ **A.** The MDR holds the instruction address, the MAR fetches the

instruction, and the PC stores the result

○ **B.** The MAR stores the program, the MDR executes it, and the PC counts completed instructions

○ **C.** The PC provides the address of the next instruction, this address is copied to the MAR, and the instruction fetched from that address is placed in the MDR

○ **D.** The PC decodes the instruction, the MAR holds data, and the MDR sends signals to the ALU

Q26 (3 marks)

A gaming company is choosing a CPU for a new gaming computer. Games typically require both fast single-core performance for game logic and multi-core performance for background tasks.
Analyse which CPU characteristics would be most important for this use case.

○ **A.** Only clock speed matters because games only use one core

○ **B.** Only the number of cores matters because more cores always means better performance

○ **C.** Only cache size matters because games require large amounts of temporary storage

○ **D.** A balance of high clock speed for fast single-threaded game logic and multiple cores for handling background processes, physics, and AI simultaneously

Q27 (3 marks)

A microcontroller manufacturer is deciding between Von Neumann and Harvard architecture for a new embedded system.
Evaluate the advantages of Harvard architecture over Von Neumann architecture for this application.

○ **A.** Harvard architecture uses separate memory and buses for instructions and data, allowing simultaneous access and eliminating the Von Neumann bottleneck

○ **B.** Harvard architecture is simpler and cheaper to manufacture

○ **C.** Harvard architecture stores programs on secondary storage, making it more secure

○ **D.** Harvard architecture uses less power because it has fewer components

Q28 (3 marks)

A software developer notices that their program runs slowly when processing a large database. The CPU has 4MB of L3 cache, and the database is 500MB.
Explain why the cache may not effectively speed up this program and suggest what could help.

○ **A.** The cache is too fast for the program; slowing down the cache would help

○ **B.** The cache only works with small files; the database should be split into 4MB files

○ **C.** The database is far larger than the cache, causing frequent cache misses as new data constantly replaces cached data; increasing cache size or optimising data access patterns could help

○ **D.** The program needs more CPU cores to utilise the cache effectively

Q29 (3 marks)

A computer architect is designing a new CPU and must decide how many registers to include.

Evaluate the trade-offs between having more registers versus fewer registers in a CPU design.

- ○ **A.** More registers always improve performance with no disadvantages
- ○ **B.** More registers reduce memory access and improve performance, but increase CPU complexity, cost, and power consumption; the optimal number balances these factors
- ○ **C.** Fewer registers are always better because they are simpler to manage
- ○ **D.** The number of registers has no effect on CPU performance

Q30 (3 marks)

A user upgrades their CPU from 2.0 GHz to 4.0 GHz but notices that some programs don't run twice as fast.
Analyse why doubling the clock speed may not result in doubled performance for all applications.

- ○ **A.** The user must also double their RAM to see any improvement
- ○ **B.** Modern programs don't use clock speed; they only use the number of cores
- ○ **C.** Clock speed has no relationship to performance; only cache size matters
- ○ **D.** Other bottlenecks like memory access speed, storage I/O, single-threaded limitations, and software efficiency can prevent the CPU from utilising its full speed

Total: **55** marks • Time: **30 min**

↑ Back to top

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